

KDTalker++: Controllable Talking Portrait Generation with Audio, Text, and Expression Editing

Chaolong Yang*

Xi'an Jiaotong-Liverpool University
Suzhou, China

Chaolong.Yang20@student.xjtlu.edu.cn

Yinuo Guo*

Duke Kunshan University
Suzhou, China

yinuo.guo@dukekunshan.edu.cn

Kai Yao

Ant Group

Hangzhou, China

jiumo.yk@antgroup.com

Yuyao Yan

Xi'an Jiaotong-Liverpool University
Suzhou, China

yuyao.yan@xjtlu.edu.cn

Jie Sun[†]

Xi'an Jiaotong-Liverpool University
Suzhou, China

Jie.Sun@xjtlu.edu.cn

Kaizhu Huang[†]

Duke Kunshan University
Suzhou, China

Kaizhu.Huang@dukekunshan.edu.cn

Abstract

This work presents KDTalker++, a real-time system for generating talking portrait videos from a single image using audio or text input. Built on a keypoint-based spatiotemporal diffusion model, it adds voice cloning, background editing, and fine-grained expression control. The demo is available at <https://kdtalker.com>. A live presentation video is available at Google drive.

CCS Concepts

• Computing methodologies → Animation.

Keywords

Talking Portrait Generation, Audio-driven, Diffusion

ACM Reference Format:

Chaolong Yang, Yinuo Guo, Kai Yao, Yuyao Yan, Jie Sun, and Kaizhu Huang. 2025. KDTalker++: Controllable Talking Portrait Generation with Audio, Text, and Expression Editing. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*, October 27–31, 2025, Dublin, Ireland. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746027.3754462>

1 Introduction

Audio-driven talking portrait generation plays a key role in enabling virtual avatars, digital humans, and multimedia content creation. Building on our previous work, KDTalker [6], which combines implicit 3D keypoints with spatiotemporal diffusion for high-fidelity generation, we further enhance the system toward practical, interactive deployment. We present KDTalker++, a demo system that extends the original framework with fine-grained control and multi-modal interactivity. New features include support for both text and audio input, adjustment of facial motion (e.g., eyes, mouth, head

pose), and background compositing, all designed to improve usability and engagement. KDTalker++ produces lip-synced, identity-preserving portrait videos from a single image, with editing and control.

2 System Overview

As illustrated in Figure 1, the KDTalker++ demo system extends our previously proposed KDTalker framework with enhanced controllability and user interactivity. The pipeline consists of four major components: (a) a keypoint-based diffusion-driven motion generation framework, (b) text-to-speech (TTS), (c) background editing, and (d) fine-grained expression and pose editing.

2.1 KDTalker Architecture

KDTalker++ is a keypoint-based spatiotemporal diffusion model synthesizing talking head videos from a single image and speech audio. The pipeline begins by extracting canonical 3D keypoints, expression deformation parameters (δ), and head pose parameters (R, t) from the reference image using a motion extractor. In parallel, the system encodes the audio or synthesized audio from TTS via a pretrained Wav2Lip [4] audio encoder to obtain frame-aligned audio features.

A reference-guided priors module initializes a structured noise sequence, which is conditioned on the extracted motion and audio features. This sequence is passed through a denoising diffusion model to generate temporally consistent motion keypoints. These keypoints are then used by a LivePortrait-based [3] renderer to produce high-fidelity video frames while preserving facial identity.

2.2 TTS Module

KDTalker++ integrates a text-to-speech (TTS) system with voice cloning capability based on IndexTTS [2]. Users can input arbitrary text and optionally provide a short reference audio clip to clone the speaker's voice and generate personalized speech audio. This speech is then fed into the audio encoder to drive the talking head animation. Voice cloning enhances identity consistency across audio and visual domains. This feature is especially powerful for content creators, educators, and digital human applications where voice style matters as much as appearance.

*Both authors contributed equally to this research.

[†]Corresponding authors.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

MM '25, Dublin, Ireland

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2035-2/2025/10

<https://doi.org/10.1145/3746027.3754462>

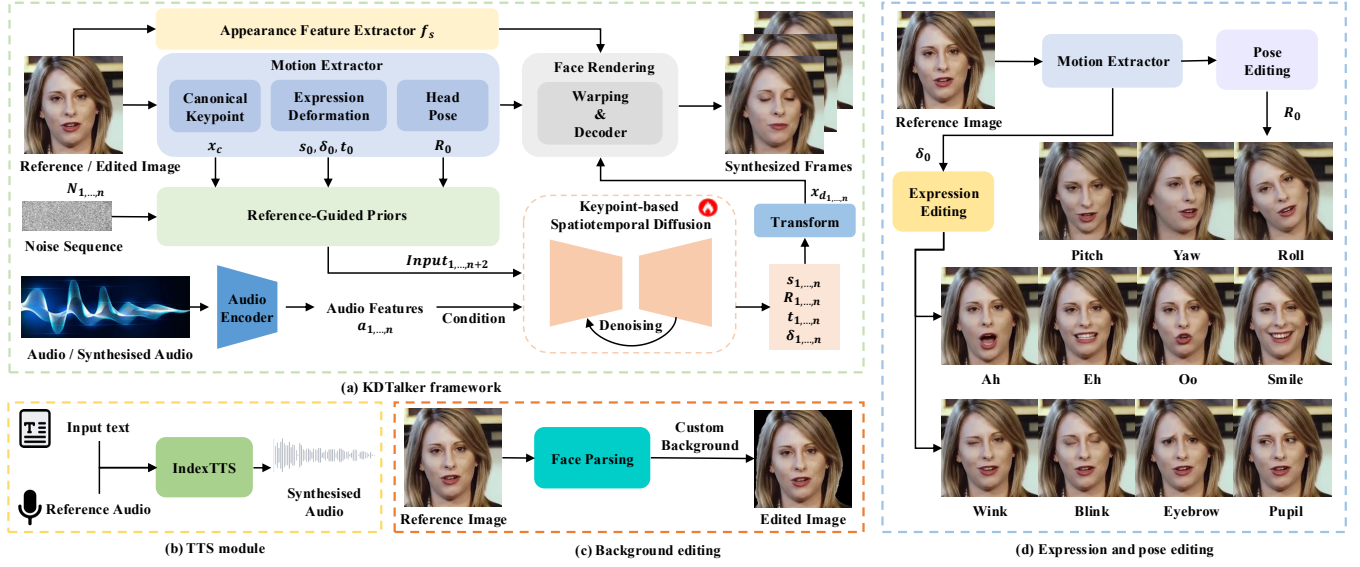


Figure 1: System overview of KDTalker++. (a) The diffusion-based keypoint generation framework. (b) Optional TTS module for typed input. (c) Background editing with facial parsing. (d) Expression and pose editing for fine-grained control.

2.3 Background Editing

KDTalker++ introduces a background editing module that allows users to replace or remove the background behind the talking head. This is achieved through a face parsing network [5] that segments the facial region from the background in the reference image. After generation, the video frames are composited with a user-chosen or uploaded background. This function makes the system suitable for practical deployment in virtual conferencing, green screen storytelling, or personalized animation creation.

2.4 Expression and Pose Editing

A key innovation in the demo is user-controllable expression and head pose editing. Users can modify the motion parameters before rendering by adjusting individual facial action units [1] (e.g., smile, blink, eyebrow down, lip shape) or head rotation (pitch, yaw, roll). These edited parameters override the predicted values from the diffusion model, allowing users to inject desired expressions or motion emphasis without retraining. This function supports creative fine-tuning, emotional personalization, and correction of undesired motion artifacts. It is especially useful in applications such as digital puppetry, interactive storytelling, or avatar-driven dialogue systems, where motion realism and consistency are critical.

Overall, the full pipeline runs at interactive speed over 25 FPS, supports audio/text input, expression control, background customization, and produces high-resolution (512×512) portrait videos with fine-grained user control.

3 Impact and Applications

KDTalker++ is a step toward practical, expressive, and interactive digital human systems. While the original KDTalker focused on model architecture and generation quality, this demo demonstrates

how such systems can be integrated into usable, real-time multimedia interfaces. The design goals of controllability, multi-modality, and user feedback enable a broad range of downstream applications:

- **Virtual Education and E-learning:** Generate talking avatars that can read aloud educational content, provide sign-language alternatives, or act as animated narrators for kids or language learners.
- **Assistive Technology:** Enable individuals with speech impairments to communicate using expressive avatars driven by TTS or symbol-based systems, enhancing accessibility and social inclusion.
- **Creative Arts and Entertainment:** Support digital puppetry, virtual concerts, or character-driven experiences where nuanced facial control is essential.

By combining state-of-the-art spatiotemporal diffusion with a real-time, modular, and user-controllable interface, KDTalker++ delivers a versatile toolkit for expressive talking portrait generation. Its design bridges the gap between research and deployment, offering value for both academic exploration and real-world multimedia applications.

4 Future Work

Though the current demo has achieved good results in talking head rendering, there is still room for improvement in both the model and engineering aspects. On the model side, it would be better if it could render full body movement instead of just stitching the head back onto a static body, which looks unnatural. We also want to explore using 3D rendering techniques [7] to support applications like game engines. On the application side, we aim to build the system using streaming to enable real-time interaction with users. We're also considering adding an LLM plug-in to make the system more interactive and useful.

Acknowledgments

The work was partially supported by the following: National Natural Science Foundation of China under No. 92370119, and 62376113.

References

- [1] AdvancedLivePortrait-WebUI. 2024. AdvancedLivePortrait-WebUI. Retrieved July 23, 2025 from <https://github.com/jhj0517/AdvancedLivePortrait-WebUI>.
- [2] Wei Deng, Siyi Zhou, Jingchen Shu, Jinchao Wang, and Lu Wang. 2025. *IndexTTS: An Industrial-Level Controllable and Efficient Zero-Shot Text-To-Speech System*. arXiv:2502.05512
- [3] Jianzhu Guo, Dingyun Zhang, Xiaoqiang Liu, Zhizhou Zhong, Yuan Zhang, Pengfei Wan, and Di Zhang. 2024. *LivePortrait: Efficient Portrait Animation with Stitching and Retargeting Control*. arXiv:2407.03168
- [4] KR Prajwal, Rudrabha Mukhopadhyay, Vinay P Namboodiri, and CV Jawahar. 2020. A lip sync expert is all you need for speech to lip generation in the wild. In *Proceedings of the 28th ACM International Conference on Multimedia (ACM MM)*. 484–492.
- [5] Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M Alvarez, and Ping Luo. 2021. SegFormer: Simple and efficient design for semantic segmentation with transformers. In *Proceedings of the Conference on Neural Information Processing Systems (NeurIPS)*, Vol. 34. 12077–12090.
- [6] Chaolong Yang, Kai Yao, Yuyao Yan, Chenru Jiang, Weiguang Zhao, Jie Sun, Guangliang Cheng, Yifei Zhang, Bin Dong, and Kaizhu Huang. 2025. *Unlock Pose Diversity: Accurate and Efficient Implicit Keypoint-based Spatiotemporal Diffusion for Audio-driven Talking Portrait*. arXiv:2503.12963
- [7] Weiguang Zhao, Chaolong Yang, Jianan Ye, Rui Zhang, Yuyao Yan, Xi Yang, Bin Dong, Amir Hussain, and Kaizhu Huang. 2025. From 2D Images to 3D Model: Weakly Supervised Multi-View Face Reconstruction with Deep Fusion. In *Proceedings of the IEEE International Conference on Multimedia and Expo (ICME)*.